

BALLARAT, AustraliaWMO: **948520**Lat: **37.5128S**Lon: **143.7911E**Elev: **436**StdP: **96.20**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.2	1.0	-1.9	3.4	10.6	-0.6	3.8	8.2	14.6	7.9	13.3	8.6	1.4	50	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	33.2	17.5	30.6	17.0	28.6	16.5	20.0	26.4	18.9	25.8	17.9	25.2	7.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	14.0	21.1	17.0	12.8	20.3	15.7	11.7	19.4	59.3	25.9	55.4	25.7	52.3	25.3	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.4	11.1	10.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-2.7	38.1	1.0	2.2	-3.5	39.7	-4.1	41.0	-4.6	42.2	-5.4	43.8		
			-2.9	22.1	1.1	2.5	-3.7	23.9	-4.3	25.4	-4.9	26.8	-5.7	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	18.5	18.5	16.2	12.8	9.6	7.4	6.7	7.3	9.1	11.3	14.1	16.2	(6)
(7)		DBStd	5.33	4.39	4.12	3.95	3.20	2.45	1.90	1.79	2.14	2.64	3.42	4.02	4.26	(7)
(8)		HDD10.0	397	0	0	1	9	38	80	102	89	47	23	6	2	(8)
(9)		HDD18.3	2390	52	44	91	169	272	328	360	343	277	219	140	96	(9)
(10)		CDD10.0	1227	264	239	193	93	24	2	1	4	21	64	128	193	(10)
(11)		CDD18.3	178	58	50	25	2	0	0	0	0	0	2	12	29	(11)
(12)		CDH23.3	2370	786	587	297	24	0	0	0	0	1	34	176	464	(12)
(13)		CDH26.7	928	355	234	95	3	0	0	0	0	0	6	47	188	(13)
(14)	Wind	WSAvg	5.2	5.7	5.6	5.3	4.7	4.7	4.9	5.4	5.3	5.3	5.2	5.4	(14)	
(15)	Precipitation	PrecAvg	681	43	39	38	53	64	60	65	74	71	68	54	47	(15)
(16)		PrecMax	996	189	240	132	136	173	170	122	138	150	193	131	138	(16)
(17)		PrecMin	343	4	0	4	9	11	14	20	13	19	12	18	3	(17)
(18)		PrecStd	136	33	42	29	33	36	30	26	25	31	35	27	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.6	35.5	33.0	26.1	19.4	14.2	13.3	16.6	21.1	27.0	30.5	35.0	(19)
(20)			MCWB	18.4	18.5	17.2	14.5	12.2	10.7	10.1	11.7	13.5	14.4	16.5	17.0	(20)
(21)		2%	DB	33.6	31.9	29.1	23.2	16.9	12.7	11.7	14.3	18.1	23.5	27.8	31.3	(21)
(22)			MCWB	17.9	17.8	16.3	13.8	12.1	10.1	9.2	10.2	12.4	14.0	15.8	16.7	(22)
(23)		5%	DB	30.4	29.6	26.6	21.2	15.2	11.7	10.7	12.6	15.9	20.7	25.1	28.4	(23)
(24)			MCWB	17.1	17.1	15.9	13.1	11.4	9.6	8.8	9.2	11.2	13.2	15.3	15.9	(24)
(25)		10%	DB	27.8	27.2	24.0	19.2	13.9	10.7	9.8	11.2	14.2	18.2	22.3	25.4	(25)
(26)			MCWB	16.4	16.8	15.4	12.6	11.0	9.2	8.2	8.5	10.5	12.3	14.5	15.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	21.6	19.5	16.4	14.5	12.2	11.4	12.3	14.7	17.0	19.5	20.2	(27)
(28)			MCDWB	25.4	28.3	25.4	22.0	15.9	12.9	12.0	15.5	19.1	22.2	25.2	25.2	(28)
(29)		2%	WB	20.1	20.0	18.1	15.2	13.2	10.9	10.0	10.7	13.1	15.3	17.7	18.4	(29)
(30)			MCDWB	26.8	26.5	24.3	19.5	15.2	11.8	11.0	13.1	16.7	20.5	23.4	25.3	(30)
(31)		5%	WB	18.8	18.8	17.1	14.3	12.2	10.1	9.2	9.8	11.9	14.0	16.4	17.2	(31)
(32)			MCDWB	27.2	25.3	23.2	18.7	14.4	11.1	10.1	11.8	15.3	19.1	22.4	25.5	(32)
(33)		10%	WB	17.6	17.8	16.1	13.4	11.3	9.5	8.5	9.0	10.8	12.9	15.2	16.1	(33)
(34)			MCDWB	25.9	24.9	22.6	17.9	13.3	10.4	9.4	10.8	13.7	17.4	21.0	23.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.5	12.7	11.4	10.1	7.5	6.3	6.1	7.3	8.8	10.5	11.4	13.2	(35)
(36)			MCDBR	18.4	17.1	15.7	14.0	9.8	8.0	7.3	9.4	11.8	15.2	16.8	18.8	(36)
(37)			MCWBR	6.0	5.8	5.5	6.2	5.8	5.6	5.1	5.8	6.9	7.7	6.8	6.9	(37)
(38)		5% WB	MCDWR	14.6	13.5	12.7	11.1	7.9	6.3	6.1	8.4	10.3	12.7	13.4	14.7	(38)
(39)			MCWBR	6.3	5.9	5.6	5.8	5.2	4.9	4.7	5.6	6.6	7.3	6.6	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.328	0.311	0.302	0.294	0.284	0.283	0.292	0.311	0.321	0.329	0.321	(40)	
(41)		taud	2.516	2.533	2.575	2.577	2.570	2.592	2.593	2.556	2.490	2.475	2.482	2.533	(41)	
(42)		Ebn at Noon	1004	982	961	909	856	838	862	907	944	978	998	1017	(42)	
(43)		Edh at Noon	111	105	92	81	70	64	67	81	99	110	114	110	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.32	6.38	5.06	3.51	2.34	1.83	2.03	2.87	4.03	5.36	6.33	7.14	(44)	
(45)		RadStd	0.45	0.33	0.29	0.22	0.13	0.14	0.13	0.19	0.28	0.45	0.41	0.41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.49	N/A	N/A	+0.87	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (40 neighbors)	+0.50	N/A	N/A	+0.86	N/A	N/A	-16	-126	+173	+59	(47)

Nomenclature: See separate page